

# 3

## Sequences of real numbers

**Note to students :** This document is currently under review. Your remarks and suggestions are welcome. Thanks.

### 1 Sequences : General definitions

This section introduces the fundamental definitions related to numerical sequences.

#### Definition 3.1 — Sequence.

A sequence in  $\mathbb{R}$  is just an ordered of an unending list of numbers.

#### Example 3.2 The lists

$$\{1, 2, 3, 4, \dots\}, \quad \{1, -3, 9, -27, \dots\}, \quad \text{and} \quad \{1, -1, 1, -1, 1, \dots\},$$

are real sequences. They can be written as

$$\{n, \quad n \in \mathbb{N} \setminus \{0\}\}, \quad \{(-3)^n, \quad n \in \mathbb{N}\}, \quad \text{and} \quad \{(-1)^n, \quad n \in \mathbb{N}\}.$$

#### Definition 3.3 — sequence.

A sequence in  $\mathbb{R}$  is a mapping

$$\begin{aligned} u : \mathbb{N} &\rightarrow \mathbb{R}, \\ n &\mapsto u(n). \end{aligned}$$

The value  $u(n)$  is often written as  $u_n$  and it is called the  $n^{th}$  term of the sequence.

The sequence  $u$  is denoted by  $(u_n)_n$  or  $\{u_n\}_n$  or  $\{u_n, \quad n \in \mathbb{N}\}$  where the subscript  $n$  is called the index.

**Remark 3.4** While many sequences are indexed from  $n = 0$ , it is equally valid to define them starting from  $n = 1$  or any other positive integer.

**Example 3.5** The sequences defined by

$$\begin{aligned} u_n &= \frac{n}{2^n}, & n \in \mathbb{N}, \\ v_n &= \frac{(-1)^n}{n}, & n \in \mathbb{N} \setminus \{0\}, \\ w_n &= \frac{1}{n-5} & n \geq 6, \end{aligned}$$

are well defined real sequences.

**Definition 3.6** — **Positive, negative and alternating sequences.**

- A sequence  $(u_n)_n$  is with positive terms if and only if there exists a rank  $n_0 \in \mathbb{N}$  such that for every  $n$  greater than or equal to  $n_0$ , we have  $u_n$  is non-negative real number.

Symbolically, this can be written with quantifiers as

$$((u_n)_n \text{ is with positive terms}) \Leftrightarrow (\exists n_0 \in \mathbb{N} : \forall n \geq n_0, u_n \geq 0);$$

- A sequence  $(u_n)_n$  is with negative terms if and only if there exists a rank  $n_0 \in \mathbb{N}$  such that for every  $n$  greater than or equal to  $n_0$ , we have  $u_n$  is non-positive real number.

Symbolically, this can be written with quantifiers as

$$((u_n)_n \text{ is with negative terms}) \Leftrightarrow (\exists n_0 \in \mathbb{N} : \forall n \geq n_0, u_n \leq 0);$$

- A sequence  $(u_n)_n$  where each successive term alternates between positive and negative values is called an alternating sequence.

Symbolically, this can be written with quantifiers as

$$((u_n)_n \text{ is an alternating sequence}) \Leftrightarrow (\forall n, u_n \times u_{n+1} < 0).$$

**Example 3.7**

- 1 The sequence  $(u_n)_{n \in \mathbb{N}}$  defined by

$$u_n = n + 1, \quad n \in \mathbb{N},$$

is with positive terms for every  $n \in \mathbb{N}$ , hence,  $n_0 = 0$ ;

- 2 The sequence  $(v_n)_{n > 0}$  defined by

$$v_n = \frac{4}{n} - 1, \quad n \in \mathbb{N} \setminus \{0\},$$

is a sequence with negative terms for every  $n > 3$ , hence,  $n_0 = 4$ ;

3 The sequence  $(w_n)_{n \geq 0}$  defined by

$$w_n = (-1)^n, \quad n \in \mathbb{N},$$

is an alternating sequence, because for every  $n$ ,  $u_{n+1} - u_n = 2(-1)^{n+1}$ .

## 2 Bounded sequences

**Definition 3.8 — Bounded from above sequences.**

A sequence  $(u_n)_n$  is said to be bounded from above if there exists a real number  $M$  such that for every  $n$ , we have  $u_n$  is less than or equal to  $M$ .

Symbolically, this can be written with quantifiers as

$$((u_n)_n \text{ is bounded from above}) \Leftrightarrow (\exists M \in \mathbb{R} : \forall n, u_n \leq M).$$

Any such number  $M$  is called an upper bound.

**Example 3.9** The sequence  $(u_n)_{n \in \mathbb{N}}$  defined by  $u_n = -n^2$ ,  $n \in \mathbb{N}$  is bounded from above by  $M = 0$ , because

$$\forall n \in \mathbb{N} : u_n \leq 0.$$

**Definition 3.10 — Bounded from below sequences.**

A sequence  $(u_n)_n$  is said to be bounded from below if there exists a real number  $m$  such that for every  $n$ , we have  $u_n$  is greater than or equal to  $m$ .

Symbolically, this can be written with quantifiers as

$$((u_n)_n \text{ is bounded from below}) \Leftrightarrow (\exists m \in \mathbb{R} : \forall n, u_n \geq m).$$

Any such number  $m$  is called a lower bound.

**Example 3.11** The sequence  $(v_n)_{n \geq 1}$  defined by  $v_n = 2^n$ ,  $n \in \mathbb{N} \setminus \{0\}$  is bounded from below by  $m = 2$ , because

$$\forall n \in \mathbb{N} \setminus \{0\} : 2 \leq v_n.$$

**Definition 3.12 — Bounded sequences.**

A sequence  $(u_n)_n$  is said to be bounded if and only if it is bounded from above and bounded from below

Symbolically, this can be written with quantifiers as

$$((u_n)_n \text{ is bounded}) \Leftrightarrow (\exists M \in \mathbb{R}, \exists m \in \mathbb{R} : \forall n, m \leq u_n \leq M),$$

or

$$((u_n)_n \text{ is bounded}) \Leftrightarrow (\exists a \in \mathbb{R}_+ \setminus \{0\} : \forall n, |u_n| \leq a).$$

**Example 3.13** The sequence  $(u_n)_{n \in \mathbb{N}}$  defined by  $u_n = \frac{n}{n+1}$ ,  $n \in \mathbb{N}$  is bounded from below by  $m = 0$  and bounded from above by  $M = 1$ . Hence,  $(u_n)_{n \in \mathbb{N}}$  is bounded, i. e.,

$$\exists m = 0 \in \mathbb{R}, \exists M = 1 \in \mathbb{R} : \forall n \in \mathbb{N}, 0 \leq u_n \leq 1.$$

**Definition 3.14 — Unbounded sequences.**

A sequence  $(u_n)_n$  is unbounded if it is not bounded.

**Example 3.15** The sequence  $(u_n)_{n \in \mathbb{N}}$  defined by  $u_n = n^2$ ,  $n \in \mathbb{N}$  is unbounded.

### 3 Monotonic sequences

**Definition 3.16 — Monotonic sequences.**

Let  $(u_n)_n$  be a sequence of real numbers. We say that  $(u_n)_n$  is

- Increasing if and only if there exists a rank  $n_0 \in \mathbb{N}$  such that for every  $n$  greater than or equal to  $n_0$ , we have  $u_n$  is less than or equal to  $u_{n+1}$ .

Symbolically, this can be written with quantifiers as

$$((u_n)_n \text{ is increasing}) \Leftrightarrow (\exists n_0 \in \mathbb{N} : \forall n \geq n_0, u_n \leq u_{n+1});$$

- Strictly increasing if and only if there exists a rank  $n_0 \in \mathbb{N}$  such that for every  $n$  greater than or equal to  $n_0$ , we have  $u_n$  is less than  $u_{n+1}$ .

Symbolically, this can be written with quantifiers as

$$((u_n)_n \text{ is strictly increasing}) \Leftrightarrow (\exists n_0 \in \mathbb{N} : \forall n \geq n_0, u_n < u_{n+1});$$

- Decreasing if and only if there exists a rank  $n_0 \in \mathbb{N}$  such that for every  $n$  greater than or equal to  $n_0$ , we have  $u_n$  is greater than or equal to  $u_{n+1}$ .

Symbolically, this can be written with quantifiers as

$$((u_n)_n \text{ is decreasing}) \Leftrightarrow (\exists n_0 \in \mathbb{N} : \forall n \geq n_0, u_n \geq u_{n+1});$$

- Strictly decreasing if and only if there exists a rank  $n_0 \in \mathbb{N}$  such that for every  $n$  greater than or equal to  $n_0$ , we have  $u_n$  is greater than  $u_{n+1}$ .

Symbolically, this can be written with quantifiers as

$$((u_n)_n \text{ is strictly decreasing}) \Leftrightarrow (\exists n_0 \in \mathbb{N} : \forall n \geq n_0, u_n > u_{n+1});$$

- Monotonic if it is either increasing or decreasing ;
- Strictly monotonic if it is either strictly increasing or strictly decreasing ;

- Constant if for every  $n$ , we have  $u_n$  is equal to  $u_{n+1}$ .

**Strategy 3.17 — Increasing / Decreasing sequences : Using difference.**

To show that a given sequence  $(u_n)_n$  is monotonic, consider the expression  $u_{n+1} - u_n$

- ▶ If  $u_{n+1} - u_n \leq 0$  for  $n \geq n_0$ , then the sequence  $(u_n)_n$  is decreasing ;
- ▶ If  $u_{n+1} - u_n \geq 0$  for  $n \geq n_0$ , then the sequence  $(u_n)_n$  is increasing.

**Strategy 3.18 — Increasing / Decreasing sequences : Using quotient.**

To show that a given sequence  $(u_n)_n$  of positive terms is monotonic, consider the expression  $\frac{u_{n+1}}{u_n}$

- ▶ If  $\frac{u_{n+1}}{u_n} \leq 1$  for  $n \geq n_0$ , then the sequence  $(u_n)_n$  is decreasing ;
- ▶ If  $\frac{u_{n+1}}{u_n} \geq 1$  for  $n \geq n_0$ , then the sequence  $(u_n)_n$  is increasing.

**Example 3.19**

- 1** Let us study the monotonicity of the sequence  $(u_n)_{n \geq 1}$  defined by

$$u_n = \sum_{k=1}^n \frac{1}{k^k}, \quad n \geq 1.$$

For  $n \geq 1$ , we have

$$\begin{aligned} u_{n+1} - u_n &= \sum_{k=1}^{n+1} \frac{1}{k^k} - \sum_{k=1}^n \frac{1}{k^k}, \\ &= \sum_{k=1}^n \frac{1}{k^k} + \frac{1}{(n+1)^{n+1}} - \sum_{k=1}^n \frac{1}{k^k}, \\ &= \frac{1}{(n+1)^{n+1}} > 0. \end{aligned}$$

As for every  $n \geq 1$ ,  $u_{n+1} - u_n > 0$ , then the sequence  $(u_n)_{n \geq 1}$  is strictly increasing (in this case  $n_0 = 1$ ).

Hence,  $(u_n)_{n \geq 1}$  is strictly monotonic sequence.

- 2** Let us check the monotonicity of the sequence  $(v_n)_{n \in \mathbb{N}}$  defined by

$$v_n = \frac{5^n}{n!}, \quad n \in \mathbb{N}.$$

As for every  $n \in \mathbb{N}$ , the sequence  $(v_n)_{n \in \mathbb{N}}$  has positive terms, then we compute  $\frac{v_{n+1}}{v_n}$ . Indeed, for  $n \in \mathbb{N}$ , we have

$$\frac{v_{n+1}}{v_n} = \frac{5}{n+1}.$$

Observe that  $\frac{5}{n+1} > 1$  for  $n = 0, \dots, 3$  and  $\frac{5}{n+1} \leq 1$  for  $n \geq 4$ . Hence, it follows that the sequence  $(v_n)_{n \in \mathbb{N}}$  is decreasing (in this case  $n_0 = 4$ ).

Consequently,  $(v_n)_{n \in \mathbb{N}}$  is monotonic sequence.

**3** Consider the sequence  $(w_n)_{n \in \mathbb{N}}$  defined by

$$w_n = (-1)^n, \quad n \in \mathbb{N}.$$

For  $n \in \mathbb{N}$ , we have

$$\begin{aligned} w_{n+1} - w_n &= (-1)^{n+1} - (-1)^n, \\ &= 2(-1)^{n+1}. \end{aligned}$$

Hence, the sequence is not monotonic (it is neither increasing nor decreasing).

## 4 Convergent sequences

### 4.1 Definitions of convergent sequences

#### Definition 3.20 — Convergent sequence.

We say that a sequence  $(u_n)_n$  is convergent or converges if there exists a real number  $\ell$  such that for every non-empty interval  $I_\ell \subset \mathbb{R}$  of mid-point  $\ell$ , there exists a rank  $n_0 \in \mathbb{N}$  for which  $u_n$  belongs to  $I_\ell$  for every  $n \geq n_0$ .

We write

$$\lim_{n \rightarrow +\infty} u_n = \ell.$$

As the mid-point of  $I_\ell$  is fixed  $\ell$ , it is necessary and sufficient to know its radius which is  $\varepsilon > 0$

Hence, the definition if the convergence become

#### Definition 3.21 — Convergent sequence : Definition of limits.

Let  $(u_n)_n$  be a numerical sequence.

We say that the sequence  $(u_n)_n$  is convergent to the real number  $\ell$  if and only if for every  $\varepsilon > 0$ , there exists a rank  $n_0 \in \mathbb{N}$ , for every  $n$ , if  $n \geq n_0$ , then  $|u_n - \ell| < \varepsilon$ .

Symbolically, this can be written with quantifiers as

$$\left( \lim_{n \rightarrow +\infty} u_n = \ell \right) \Leftrightarrow \left( \forall \varepsilon > 0, \exists n_0 \in \mathbb{N}, \forall n : n \geq n_0 \Rightarrow |u_n - \ell| < \varepsilon \right).$$

#### Remark 3.22

- According to the definition of convergent sequences, the sequence  $(u_n)_n$  converges to the limit  $\ell \in \mathbb{R}$  is equivalent to say that there exists a rank  $n_0$  such that all element of  $(u_n)_n$  belong to the interval  $]\ell - \varepsilon, \ell + \varepsilon[$  ;

- If  $\ell = 0$ , then the sequence  $(u_n)_n$  is said to be null sequence ;
- Every constant sequence  $(u_n)_n$  defined by  $u_n = c \in \mathbb{R}$  converges to  $c$ .

**Example 3.23** Using the definition of the convergence, show that

$$\lim_{n \rightarrow +\infty} \frac{3n-4}{2n+1} = \frac{3}{2}.$$

We are asked to establish that

$$\forall \varepsilon > 0, \exists n_0 \in \mathbb{N}, \forall n \in \mathbb{N} : n \geq n_0 \Rightarrow \left| \frac{3n-4}{2n+1} - \frac{3}{2} \right| < \varepsilon.$$

Let  $n \in \mathbb{N}$ , we have

$$\begin{aligned} \left| \frac{3n-4}{2n+1} - \frac{3}{2} \right| &= \left| \frac{-11}{4n+2} \right|, \\ &= \frac{11}{4n+2}. \end{aligned}$$

Fix  $\varepsilon > 0$ , for  $n \in \mathbb{N}$ , we have

$$\begin{aligned} \left| \frac{3n-4}{2n+1} - \frac{3}{2} \right| < \varepsilon &\Rightarrow \frac{11}{4n+2} < \varepsilon, \\ &\Rightarrow n > \frac{11}{4\varepsilon} - \frac{1}{2}. \end{aligned}$$

Therefore, it is enough to take  $n_0 = \left\lfloor \frac{11}{4\varepsilon} - \frac{1}{2} \right\rfloor + 1 \in \mathbb{N}$ .

Consequently,

$$\forall \varepsilon > 0, \exists n_0 = \left\lfloor \frac{11}{4\varepsilon} - \frac{1}{2} \right\rfloor + 1 \in \mathbb{N}, \forall n \in \mathbb{N} : n \geq n_0 \Rightarrow \left| \frac{3n-4}{2n+1} - \frac{3}{2} \right| < \varepsilon,$$

that is equivalent to

$$\lim_{n \rightarrow +\infty} \frac{3n-4}{2n+1} = \frac{3}{2}.$$

**Example 3.24** Using the definition of the convergence, show that

$$\lim_{n \rightarrow +\infty} \left( -\frac{1}{2} \right)^n = 0.$$

We aim to prove that

$$\forall \varepsilon > 0, \exists n_0 \in \mathbb{N}, \forall n \in \mathbb{N} : n \geq n_0 \Rightarrow \left| \left( -\frac{1}{2} \right)^n \right| < \varepsilon.$$

Let  $n \in \mathbb{N}$ , we have

$$\begin{aligned} \left| \left( -\frac{1}{2} \right)^n \right| &= \left( \left| -\frac{1}{2} \right| \right)^n, \\ &= \frac{1}{2^n}. \end{aligned}$$

Fix  $\varepsilon > 0$ , for  $n \in \mathbb{N}$ , we have

$$\begin{aligned} \left| \left( -\frac{1}{2} \right)^n \right| < \varepsilon &\Rightarrow \frac{1}{2^n} < \varepsilon, \\ &\Rightarrow n > \frac{-\ln(\varepsilon)}{\ln(2)}. \end{aligned}$$

Therefore, it suffices to choose  $n_0 = \left\lfloor \frac{-\ln(\varepsilon)}{\ln(2)} \right\rfloor + 1 \in \mathbb{N}$ .

Consequently,

$$\forall \varepsilon > 0, \exists n_0 = \left\lfloor \frac{-\ln(\varepsilon)}{\ln(2)} \right\rfloor + 1 \in \mathbb{N}, \forall n \in \mathbb{N} : n \geq n_0 \Rightarrow \left| \left( -\frac{1}{2} \right)^n \right| < \varepsilon,$$

that is equivalent to

$$\lim_{n \rightarrow +\infty} \left( -\frac{1}{2} \right)^n = 0.$$

**Strategy 3.25 — How to use the definition of convergent sequence.**

To show that for a given sequence  $(u_n)_n$ ,  $\lim_{n \rightarrow +\infty} u_n = \ell$ :

- ▶ Consider the expression  $|u_n - \ell|$ ;
- ▶ Solve the inequality,  $|u_n - \ell| < \varepsilon$  to find a number  $n_0 \in \mathbb{N}$  generally depending on  $\varepsilon$ .

## 4.2 Theoretical results on convergent sequences

Convergent sequences are governed by various theorems. In the following, we highlight the most essential.

### 4.2.1 Some properties of convergent sequences

**Theorem 3.26 — Uniqueness of the limit.**

If the sequence  $(u_n)_n$  converges to  $\ell \in \mathbb{R}$ , then  $\ell$  is uniquely determined.

In mathematical notation, Theorem 3.26 can be expressed as

$$\lim_{n \rightarrow +\infty} u_n = \ell \Rightarrow \exists ! \ell \in \mathbb{R}.$$

**Theorem 3.27 — Boundedness Theorem.**

Every convergent sequence of real numbers is bounded

In mathematical notation, Theorem 3.27 can be expressed as

$$(\forall \varepsilon > 0, \exists n_0 \in \mathbb{N}, \forall n : n \geq n_0 \Rightarrow |u_n - \ell| < \varepsilon) \Rightarrow (\exists m, M \in \mathbb{R}, \forall n : m \leq u_n \leq M).$$

**Example 3.28** It is obvious that

$$\lim_{n \rightarrow +\infty} \frac{1}{n} = 0,$$

hence, the sequence defined by  $u_n = \frac{1}{n}$ ,  $n \geq 1$  is bounded, and we have

$$\forall n \in \mathbb{N} \setminus \{0\} : 0 < \frac{1}{n} \leq n.$$

**Remark 3.29** A bounded sequence is not necessarily convergent. For instance, the sequence  $(u_n)_n$  defined by  $u_n = (-1)^n$ ,  $n \in \mathbb{N}$  is bounded, i. e.,

$$\forall n \in \mathbb{N} : -1 \leq u_n \leq 1,$$

but it does not converge.

**Lemma 3.30** If a numerical sequence  $(u_n)_n$  converges to  $\ell \neq 0$ , then there exists a rank  $n_0$  such that for every  $n$

$$n \geq n_0 \Rightarrow |u_n| > \frac{|\ell|}{2}.$$

### 4.2.2 Combination rules for convergent sequences

**Theorem 3.31** Let  $(u_n)_n$  be a convergent sequence with  $\lim_{n \rightarrow +\infty} u_n = \ell$ . Then we have

**(a) Absolute value rule :** The sequence  $(|u_n|)_n$  converges, and we have

$$\lim_{n \rightarrow +\infty} |u_n| = |\ell|;$$

**(b) Multiplication rule :** For every  $\lambda \in \mathbb{R}$ , the sequence  $(\lambda u_n)_n$  is convergent, and we have

$$\lim_{n \rightarrow +\infty} (\lambda u_n) = \lambda \left( \lim_{n \rightarrow +\infty} u_n \right) = \lambda \ell;$$

**(c) Reciprocal rule :** The sequence  $\left(\frac{1}{u_n}\right)_n$  converges, and we have

$$\lim_{n \rightarrow +\infty} \frac{1}{u_n} = \frac{1}{\ell},$$

provided that  $\ell \neq 0$ .

**Theorem 3.32** Let  $(u_n)_n$  and  $(v_n)_n$  be two convergent sequences with  $\lim_{n \rightarrow +\infty} u_n = \ell_u$  and  $\lim_{n \rightarrow +\infty} v_n = \ell_v$ . Then we have

**(a) Sum rule :** The sequence  $(u_n + v_n)_n$  converges and

$$\lim_{n \rightarrow +\infty} (u_n + v_n) = \lim_{n \rightarrow +\infty} u_n + \lim_{n \rightarrow +\infty} v_n = \ell_u + \ell_v;$$

**(b) Product rule :** The sequence  $(u_n \times v_n)_n$  converges and

$$\lim_{n \rightarrow +\infty} (u_n \times v_n) = \left( \lim_{n \rightarrow +\infty} u_n \right) \times \left( \lim_{n \rightarrow +\infty} v_n \right) = \ell_u \times \ell_v;$$

**(c) Quotient rule :** The sequence  $\left(\frac{u_n}{v_n}\right)_n$  converges and

$$\lim_{n \rightarrow +\infty} \frac{u_n}{v_n} = \frac{\lim_{n \rightarrow +\infty} u_n}{\lim_{n \rightarrow +\infty} v_n} = \frac{l_u}{l_v},$$

provided that  $l_v \neq 0$ .

### 4.2.3 Convergent sequences and order relation

**Proposition 3.33** Let  $(u_n)_n$  be a numerical sequence that converges to the limit  $\ell$  and there exists a rank  $n_0 \in \mathbb{N}$  such that for every  $n \geq n_0$ ,  $a < u_n < b$ , then

$$a \leq \ell \leq b.$$

**Corollary 3.34** Let  $(u_n)_n$  be a convergent sequence to a limit  $\ell \in \mathbb{R}$ .

If there exists a rank  $n_0 \in \mathbb{N}$  such that for every  $n \geq n_0$ ,  $u_n < a$  (respectively  $u_n > a$ ), then  $\ell \leq a$  (respectively  $\ell \geq a$ ).

**Remark 3.35** Proposition 3.33 and Corollary 3.34 hold even if  $a \leq u_n \leq b$  and  $u_n \leq a$  (respectively  $u_n \geq a$ ).

#### Example 3.36

**1** Let  $(u_n)_n$  be a sequence defined by  $u_n = \frac{n}{n+1}$ ,  $n \in \mathbb{N}$ .

It is clear that for every  $n \in \mathbb{N}$ ,  $u_n > 1$ , however,

$$\lim_{n \rightarrow +\infty} u_n = 1;$$

**2** Let  $(v_n)_n$  be a sequence defined by  $v_n = \frac{1}{n}$ ,  $n \in \mathbb{N} \setminus \{0\}$ .

It is clear that for every  $n \geq 1$ ,  $v_n > 0$ , nevertheless,

$$\lim_{n \rightarrow +\infty} v_n = 0.$$

#### Theorem 3.37 — Limit inequality rule.

Let  $(u_n)_n$  and  $(v_n)_n$  be two convergent sequences.

If  $\lim_{n \rightarrow +\infty} u_n = \ell_u$  and  $\lim_{n \rightarrow +\infty} v_n = \ell_v$  and  $u_n < v_n$ , then

$$\ell_u \leq \ell_v.$$

**Example 3.38** Consider the sequences  $(u_n)_n$  and  $(v_n)_n$  defined respectively by

$$u_n = \frac{1}{n}, \quad \text{and} \quad v_n = \frac{2}{n}, \quad n \geq 1.$$

It is obvious that

$$\forall n \in \mathbb{N} \setminus \{0\} : u_n < v_n,$$

however,

$$\lim_{n \rightarrow +\infty} u_n = 0 = \lim_{n \rightarrow +\infty} v_n.$$

**Example 3.39** Using the definition of convergent sequence, let us establish that

$$\lim_{n \rightarrow +\infty} \frac{n^2 - n + 2}{3n^2 + 2n - 4} = \frac{1}{3}.$$

We aim to show that

$$\forall \varepsilon > 0, \exists n_0 \in \mathbb{N}, \forall n \in \mathbb{N} \setminus \{0\} : n \geq n_0 \Rightarrow \left| \frac{n^2 - n + 2}{3n^2 + 2n - 4} - \frac{1}{3} \right| < \varepsilon.$$

For  $n \geq 1$ , we have

$$\begin{aligned} \left| \frac{n^2 - n + 2}{3n^2 + 2n - 4} - \frac{1}{3} \right| &= \frac{5n - 10}{9n^2 + 6n - 12}, \\ &< \frac{5n}{8n^2}, \quad \text{because, } \forall n \geq 2 \text{ we have } 5n - 10 \leq 5n \text{ and } 9n^2 + 6n - 12 > 8n^2, \\ &< \frac{1}{n}, \quad \text{because, } \frac{5}{8} < 1. \end{aligned}$$

Hence, for  $\varepsilon > 0$  and  $n \geq 2$ , we have

$$\left| \frac{n^2 - n + 2}{3n^2 + 2n - 4} - \frac{1}{3} \right| < \frac{1}{n} < \varepsilon \Rightarrow n > \frac{1}{\varepsilon},$$

therefore, it is enough to choose  $n_0 = \max\left(2, \left\lceil \frac{1}{\varepsilon} \right\rceil + 1\right) \in \mathbb{N}$ .

Finally,

$$\forall \varepsilon > 0, \exists n_0 = \max\left(2, \left\lceil \frac{1}{\varepsilon} \right\rceil + 1\right) \in \mathbb{N}, \forall n \in \mathbb{N} \setminus \{0\} : n \geq n_0 \Rightarrow \left| \frac{n^2 - n + 2}{3n^2 + 2n - 4} - \frac{1}{3} \right| < \varepsilon,$$

that is

$$\lim_{n \rightarrow +\infty} \frac{n^2 - n + 2}{3n^2 + 2n - 4} = \frac{1}{3}.$$

**Strategy 3.40** — How to use the definition of convergent sequence together with the order relation.

To show that for a given sequence  $(u_n)_n$ ,  $\lim_{n \rightarrow +\infty} u_n = \ell$  :

- ▶ Try to bound from above the expression  $|u_n - \ell|$  by a new null sequence  $(v_n)_n$ , i. e.  $|u_n - \ell| \leq v_n$  with  $\lim_{n \rightarrow +\infty} v_n = 0$  ;
- ▶ Solve the inequality,  $v_n < \varepsilon$  to find a number  $n_0 \in \mathbb{N}$  generally depending on  $\varepsilon$ .

**Theorem 3.41** — Squeeze theorem.

Let  $(u_n)_n$ ,  $(v_n)_n$  and  $(w_n)_n$  be numerical sequences satisfying there exists a rank  $n_0 \in \mathbb{N}$  such that for

every  $n \geq n_0$

$$v_n < u_n < w_n.$$

If  $\lim_{n \rightarrow +\infty} v_n = \ell = \lim_{n \rightarrow +\infty} w_n$ , then  $(u_n)_n$  converges to  $\ell$ , i. e.,

$$\lim_{n \rightarrow +\infty} u_n = \ell.$$

**Proposition 3.42** Let  $(u_n)_n$  and  $(v_n)_n$  be two numerical sequences.

If the sequence  $(u_n)_n$  is bounded and the sequence  $(v_n)_n$  is null sequence (i. e.,  $\lim_{n \rightarrow +\infty} v_n = 0$ ), then the sequence  $(u_n \times v_n)_n$  is convergent and

$$\lim_{n \rightarrow +\infty} (u_n \times v_n) = 0.$$

**Example 3.43** Let us evaluate  $\lim_{n \rightarrow +\infty} \frac{\sin(n)}{n}$ .

It clear that

$$\lim_{n \rightarrow +\infty} \frac{\sin(n)}{n} = \lim_{n \rightarrow +\infty} \sin(n) \times \frac{1}{n}.$$

As the sequence  $(\sin(n))_n$  is bounded sequence (i. e.,  $\forall n \in \mathbb{N} : -1 \leq \sin(n) \leq 1$ ) and  $\lim_{n \rightarrow +\infty} \frac{1}{n} = 0$ , then according to the proposition

$$\lim_{n \rightarrow +\infty} \frac{\sin(n)}{n} = 0.$$

#### 4.2.4 Convergence through boundedness and monotonicity

**Theorem 3.44 — Monotone convergence theorem.**

Let  $(u_n)_n$  be a numerical sequence.

If the sequence  $(u_n)_n$  is monotone, then it converges if and only if it is bounded.

Furthermore,

- (a) If the sequence  $(u_n)_n$  is increasing and bounded above, then it converges and tends to its least upper bound.

In mathematical notation, it can be expressed as

$$\left[ \left( \exists n_0 \in \mathbb{N} : \forall n \geq n_0, u_n \leq u_{n+1} \right) \wedge \left( \exists M \in \mathbb{R} : \forall n, u_n \leq M \right) \right] \Rightarrow \left[ \lim_{n \rightarrow +\infty} u_n = \sup \{u_n\} \right];$$

- (b) If the sequence  $(u_n)_n$  is decreasing and bounded below, then it converges and tends to its greatest lower bound.

In mathematical notation, it can be expressed as

$$\left[ \left( \exists n_0 \in \mathbb{N} : \forall n \geq n_0, u_n \geq u_{n+1} \right) \wedge \left( \exists m \in \mathbb{R} : \forall n, m \leq u_n \right) \right] \Rightarrow \left[ \lim_{n \rightarrow +\infty} u_n = \inf \{u_n\} \right].$$

**Example 3.45** Let us show that the sequence  $(u_n)_n$  defined by

$$u_n = \sum_{k=1}^n \frac{1}{n+k},$$

is convergent.

First, let us show that the sequence  $(u_n)_n$  is monotonic.

For  $n \in \mathbb{N} \setminus \{0\}$ , we have

$$\begin{aligned} u_{n+1} - u_n &= \sum_{k=1}^{n+1} \frac{1}{n+1+k} - \sum_{k=1}^n \frac{1}{n+k}, \\ &= \frac{1}{2n+1} + \frac{1}{2n+2} - \frac{1}{n+1}, \\ &= \frac{1}{2(2n+1)(n+1)} > 0, \end{aligned}$$

therefore, the sequence  $(u_n)_n$  is increasing for every  $n \geq 1$ .

Now, let us show that the sequence  $(u_n)_n$  is bounded.

For  $n \geq 1$ , we have

$$\begin{aligned} 1 \leq k \leq n &\Rightarrow n+1 \leq n+k \leq 2n, \\ &\Rightarrow \frac{1}{2n} \leq \frac{1}{n+k} \leq \frac{1}{n+1}, \\ &\Rightarrow \frac{1}{2n} \sum_{k=1}^n 1 \leq \sum_{k=1}^n \frac{1}{n+k} \leq \frac{1}{n+1} \sum_{k=1}^n 1, \\ &\Rightarrow \frac{n}{2n} \leq \sum_{k=1}^n \frac{1}{n+k} \leq \frac{n}{n+1}, \end{aligned}$$

which gives

$$\frac{1}{2} \leq u_n < 1.$$

Thus, the sequence  $(u_n)_n$  is bounded above by  $M = 1$ .

Consequently, the sequence  $(u_n)_n$  is convergent and  $\frac{1}{2} \leq \lim_{n \rightarrow +\infty} u_n \leq 1$ .

**Example 3.46** Let us show that the sequence  $(v_n)_n$  defined by

$$v_n = \frac{1 \times 3 \times 5 \times \cdots \times (2n-1)}{2 \times 4 \times 6 \times \cdots \times (2n)},$$

is convergent.

First, let us show that the sequence  $(v_n)_n$  is monotonic.

It is clear that for every  $n \geq 1$ ,  $v_n > 0$ , hence, for  $n \in \mathbb{N} \setminus \{0\}$ , we have

$$\begin{aligned} \frac{v_{n+1}}{v_n} &= \frac{v_n \times \frac{2n+1}{2n+2}}{v_n}, \\ &= \frac{2n+1}{2n+2}, \\ &< 1, \quad \text{because } 2n+1 < 2n+2, \end{aligned}$$

therefore, the sequence  $(v_n)_n$  is decreasing for every  $n \geq 1$ .

Now, let us show that the sequence  $(v_n)_n$  is bounded.

As for every  $n \geq 1$ ,  $v_n > 0$ , then the sequence  $(v_n)_n$  is bounded below by  $m = 0$ .

Thus, the sequence  $(v_n)_n$  is convergent and  $\lim_{n \rightarrow +\infty} v_n \geq 0$ .

## 5 Subsequences

### Definition 3.47 — Subsequences.

Let  $(u_n)_n$  be a numerical sequence of real numbers, and let

$$\begin{aligned} \varphi : \mathbb{N} &\rightarrow \mathbb{N}, \\ n &\mapsto \varphi(n), \end{aligned}$$

be a strictly increasing function.

The sequence  $(u_{\varphi(n)})_n$  is called a subsequence of  $(u_n)_n$ .

**Example 3.48** The sequences  $(v_n)_n$  and  $(w_n)_n$  defined respectively by

$$v_n = \frac{2}{2n+1} \quad \text{and} \quad w_n = \frac{2}{n}, \quad n \geq 1$$

are subsequences of the sequence  $(u_n)_n$  defined by

$$u_n = \frac{(-1)^n + 3}{n}, \quad n \geq 1.$$

Indeed, for

$$\begin{aligned} \varphi_1 : \mathbb{N} &\rightarrow \mathbb{N}, & \text{and} & & \varphi_2 : \mathbb{N} &\rightarrow \mathbb{N}, \\ \mapsto \varphi_1(n) &= 2n+1 & & & \mapsto \varphi_2(n) &= 2n, \end{aligned}$$

we get

$$u_{\varphi_1(n)} = \frac{(-1)^{2n+1} + 3}{2n+1} = v_n,$$

and

$$u_{\varphi_2(n)} = \frac{(-1)^{2n} + 3}{2n} = w_n,$$

### Remark 3.49

- The fact that the function  $\varphi$  is strictly increasing, then we have

$$\forall n \in \mathbb{N} : \varphi(n) \geq n;$$

- The odd  $(u_{2n+1})_n$  and even  $(u_{2n})_n$  subsequences are very useful.

### Theorem 3.50 — Inheritance property of subsequences.

If a sequence  $(u_n)_n$  of real numbers converges to the real number  $\ell$ , then all subsequences  $(u_{\varphi(n)})_n$  converge to  $\ell$ .

**Example 3.51** The sequence  $(u_n)_n$  defined by  $u_n = \frac{1}{n}$ ,  $n \in \mathbb{N} \setminus \{0\}$  converges to 0. Hence, the subsequences  $(u_{4n})_n$ ,  $(u_{4n+1})_n$ ,  $(u_{4n+2})_n$ , and  $(u_{4n+3})_n$  defined for every respectively by

$$u_{4n} = \frac{1}{4n}, \quad \text{for } n \in \mathbb{N} \setminus \{0\},$$

and for  $n \in \mathbb{N}$

$$u_{4n+1} = \frac{1}{4n+1}, \quad u_{4n+2} = \frac{1}{4n+2}, \quad \text{and} \quad u_{4n+3} = \frac{1}{4n+3},$$

converge also to 0.

**Theorem 3.52**

If the odd  $(u_{2n+1})_n$  and the even  $(u_{2n})_n$  subsequences of the sequence  $(u_n)_n$  both tend to the same limit  $\ell \in \mathbb{R}$ , then

$$\lim_{n \rightarrow +\infty} u_n = \ell.$$

**Example 3.53** Let  $(u_n)_n$  be the sequence defined by

$$u_n = (3 + (-1)^n)^{\frac{1}{n}}, \quad n \geq 1.$$

The subsequences

$$u_{2n} = 4^{\frac{1}{2n}}, \quad n \geq 1,$$

and

$$u_{2n+1} = 2^{\frac{1}{2n+1}}, \quad n \geq 0,$$

converge both to 1. Thus

$$\lim_{n \rightarrow +\infty} u_n = 1.$$

**Theorem 3.54 — Bolzano-Weiestrass Theorem.**

Every bounded sequence has convergent subsequences.

**Example 3.55** The sequence  $(u_n)_n$  given by  $u_n = (-1)^n$ ,  $n \in \mathbb{N}$ , is bounded. Its subsequences  $u_{2n} = 1$  and  $u_{2n+1} = -1$  are both convergent but not to the same limit.

**Definition 3.56 — Cluster points.**

A cluster point of a numerical sequence  $(u_n)_n$  of real numbers is any real number which is the limit of the subsequences of  $(u_n)_n$ .

**Example 3.57** Let us consider the sequence  $(u_n)_n$  defined by

$$u_n = \cos\left(\frac{n\pi}{3}\right), \quad n \in \mathbb{N}.$$

We have

$$\begin{aligned}
 u_{6n} &= \cos(2n\pi) = 1 \Rightarrow \lim_{n \rightarrow +\infty} u_{6n} = 1, \\
 u_{6n+1} &= \cos\left(2n\pi + \frac{\pi}{3}\right) = \cos\left(\frac{\pi}{3}\right) = \frac{1}{2} \Rightarrow \lim_{n \rightarrow +\infty} u_{6n+1} = \frac{1}{2}, \\
 u_{6n+2} &= \cos\left(2n\pi + \frac{2\pi}{3}\right) = \cos\left(\frac{2\pi}{3}\right) = -\frac{1}{2} \Rightarrow \lim_{n \rightarrow +\infty} u_{6n+2} = -\frac{1}{2}, \\
 u_{6n+3} &= \cos(2n\pi + \pi) = \cos(\pi) = -1 \Rightarrow \lim_{n \rightarrow +\infty} u_{6n+3} = -1, \\
 u_{6n+4} &= \cos\left(2n\pi + \frac{4\pi}{3}\right) = \cos\left(\frac{4\pi}{3}\right) = -\frac{1}{2} \Rightarrow \lim_{n \rightarrow +\infty} u_{6n+4} = -\frac{1}{2}, \\
 u_{6n+5} &= \cos\left(2n\pi + \frac{5\pi}{3}\right) = \cos\left(\frac{5\pi}{3}\right) = \frac{1}{2} \Rightarrow \lim_{n \rightarrow +\infty} u_{6n+5} = \frac{1}{2}.
 \end{aligned}$$

Therefore, the numbers  $-1, -\frac{1}{2}, \frac{1}{2}$  and  $1$  are the cluster points of the sequence  $(u_n)_n$ .

## 6 Sequences which tend to infinity

### 6.1 Definitions and examples

#### Definition 3.58 — Blowing up to positive infinity.

Let  $(u_n)_n$  be a numerical sequence of real numbers.

We say that the sequence  $(u_n)_n$  tends to positive infinity as  $n$  approaches to positive infinity if for each positive number  $A$  large enough, there exists a natural number  $n_A$  such that for every  $n$ , if  $n$  is greater than or equal to  $n_A$ , then  $u_n$  is greater than  $A$ .

In mathematical notation, it can be expressed as

$$\left(\lim_{n \rightarrow +\infty} u_n = +\infty\right) \Leftrightarrow \left(\forall A > 0, \exists n_A \in \mathbb{N}, \forall n : n \geq n_A \Rightarrow u_n > A\right).$$

**Remark 3.59**  $\lim_{n \rightarrow +\infty} u_n = +\infty$  means that for a rank  $n_A$ , all element of the sequence  $(u_n)_n$  belong to  $]A, +\infty[$ .

**Example 3.60** Using the definition of the limit, show that

$$\lim_{n \rightarrow +\infty} \frac{n^3}{2} = +\infty.$$

We are asked to establish that

$$\left(\lim_{n \rightarrow +\infty} \frac{n^3}{2} = +\infty\right) \Leftrightarrow \left(\forall A > 0, \exists n_A \in \mathbb{N}, \forall n \in \mathbb{N} : n \geq n_A \Rightarrow \frac{n^3}{2} > A\right).$$

Fix  $A > 0$  (large enough), for  $n \in \mathbb{N}$ , we have

$$\begin{aligned}
 \frac{n^3}{2} > A &\Rightarrow n^3 > 2A, \\
 &\Rightarrow n > \sqrt[3]{2A},
 \end{aligned}$$

hence, it is enough to take  $n_A = \left\lceil \sqrt[3]{2A} \right\rceil + 1 \in \mathbb{N}$ .

Therefore,

$$\forall A > 0, \exists n_A = \left\lfloor \sqrt[3]{2A} \right\rfloor + 1 \in \mathbb{N}, \forall n \in \mathbb{N} : n \geq n_A \Rightarrow \frac{n^3}{2} > A,$$

that is equivalent to

$$\lim_{n \rightarrow +\infty} \frac{n^3}{2} = +\infty.$$

**Example 3.61** Using the definition of the limit, show that

$$\lim_{n \rightarrow +\infty} (n! + 10^n) = +\infty.$$

We need to establish that

$$\left( \lim_{n \rightarrow +\infty} (n! + 10^n) = +\infty \right) \Leftrightarrow \left( \forall A > 0, \exists n_A \in \mathbb{N}, \forall n \in \mathbb{N} : n \geq n_A \Rightarrow (n! + 10^n) > A \right).$$

It is obvious that

$$\forall n \in \mathbb{N} : n! + 10^n > 10^n, \quad (10^n \text{ dominates } n!).$$

Fix  $A > 0$  (large enough), for  $n \in \mathbb{N}$ , we have

$$\begin{aligned} n! + 10^n > 10^n > A &\Rightarrow n \ln(10) > \ln(A), \\ &\Rightarrow n > \frac{\ln(A)}{\ln(10)}, \end{aligned}$$

we can choose  $n_A = \left\lfloor \frac{\ln(A)}{\ln(10)} \right\rfloor + 1 \in \mathbb{N}$ .

Thus,

$$\forall A > 0, \exists n_A = \left\lfloor \frac{\ln(A)}{\ln(10)} \right\rfloor + 1 \in \mathbb{N}, \forall n \in \mathbb{N} : n \geq n_A \Rightarrow n! + 10^n > A,$$

that is equivalent to

$$\lim_{n \rightarrow +\infty} (n! + 10^n) = +\infty.$$

**Definition 3.62 — Blowing up to negative infinity.**

Let  $(u_n)_n$  be a numerical sequence of real numbers.

We say that the sequence  $(u_n)_n$  tends to negative infinity as  $n$  approaches to positive infinity if for each negative number  $B$ , there exists a natural number  $n_B$  such that for every  $n$ , if  $n$  is greater than or equal to  $n_B$ , then  $u_n$  is less than  $B$ .

In mathematical notation, it can be expressed as

$$\left( \lim_{n \rightarrow +\infty} u_n = -\infty \right) \Leftrightarrow \left( \forall B < 0, \exists n_B \in \mathbb{N}, \forall n : n \geq n_B \Rightarrow u_n < B \right).$$

**Remark 3.63**  $\lim_{n \rightarrow +\infty} u_n = -\infty$  means that for a rank  $n_B$ , all element of the sequence  $(u_n)_n$  belong to  $] -\infty, B[$ .

**Example 3.64** Using the definition of the limit, show that

$$\lim_{n \rightarrow +\infty} (1 - 2n^2) = -\infty.$$

We are required to prove that

$$\left( \lim_{n \rightarrow +\infty} (1 - 2n^2) = -\infty \right) \Leftrightarrow \left( \forall B < 0, \exists n_B \in \mathbb{N}, \forall n \in \mathbb{N} : n \geq n_B \Rightarrow (1 - 2n^2) < B \right).$$

Fix  $B < 0$ , for  $n \in \mathbb{N}$ , we have

$$\begin{aligned} 1 - 2n^2 < B &\Rightarrow 2n^2 > -B + 1, \\ &\Rightarrow n > \sqrt{\frac{-B + 1}{2}}, \end{aligned}$$

hence, it is enough to take  $n_B = \left\lfloor \sqrt{\frac{-B + 1}{2}} \right\rfloor + 1 \in \mathbb{N}$ .

Therefore,

$$\forall B < 0, \exists n_B = \left\lfloor \sqrt{\frac{-B + 1}{2}} \right\rfloor + 1 \in \mathbb{N}, \forall n \in \mathbb{N} : n \geq n_B \Rightarrow 1 - 2n^2 < B,$$

that is equivalent to

$$\lim_{n \rightarrow +\infty} (1 - 2n^2) = -\infty.$$

**Example 3.65** Using the definition of the limit, show that

$$\lim_{n \rightarrow +\infty} \frac{1 - 2n^2}{2n + 3} = -\infty.$$

We have to show that

$$\left( \lim_{n \rightarrow +\infty} \frac{1 - 2n^2}{2n + 3} = -\infty \right) \Leftrightarrow \left( \forall B < 0, \exists n_B \in \mathbb{N}, \forall n \in \mathbb{N} : n \geq n_B \Rightarrow \frac{1 - 2n^2}{2n + 3} < B \right).$$

For  $n \geq 1$ , we have

$$\frac{1 - 2n^2}{2n + 3} > 1 - 2n^2, \quad \left( \text{because } 1 - 2n^2 < 0 \text{ and } \frac{1}{2n + 3} < 1 \right).$$

Let  $B < 0$ , for  $n \in \mathbb{N} \setminus \{0\}$ , we have

$$\begin{aligned} 1 - 2n^2 < \frac{1 - 2n^2}{2n + 3} < B &\Rightarrow 1 - 2n^2 < B, \\ &\Rightarrow n > \sqrt{\frac{-B + 1}{2}}, \end{aligned}$$

it suffices to choose  $n_B = \left\lfloor \sqrt{\frac{-B + 1}{2}} \right\rfloor + 1 \in \mathbb{N}$ .

Hence,

$$\forall B < 0, \exists n_B = \min \left( 1, \left\lfloor \sqrt{\frac{-B + 1}{2}} \right\rfloor + 1 \right) \in \mathbb{N}, \forall n \in \mathbb{N} : n \geq n_B \Rightarrow \frac{1 - 2n^2}{2n + 3} < B,$$

that is equivalent to

$$\lim_{n \rightarrow +\infty} \frac{1 - 2n^2}{2n + 3} = -\infty.$$

## 6.2 Combination rules for infinite limits

**Theorem 3.66** Let  $(u_n)_n$  and  $(v_n)_n$  be two sequences of real numbers , then

**(a) Multiplication rule :**

- If  $\lim_{n \rightarrow +\infty} u_n = +\infty$  and  $\lambda > 0$ , then  $\lim_{n \rightarrow +\infty} (\lambda u_n) = +\infty$  ;
- If  $\lim_{n \rightarrow +\infty} u_n = -\infty$  and  $\lambda > 0$ , then  $\lim_{n \rightarrow +\infty} (\lambda u_n) = -\infty$  ;

**(b) Sum rule :**

- If  $\lim_{n \rightarrow +\infty} u_n = +\infty$  and  $\lim_{n \rightarrow +\infty} v_n = +\infty$ , then  $\lim_{n \rightarrow +\infty} (u_n + v_n) = +\infty$  ;
- If  $\lim_{n \rightarrow +\infty} u_n = -\infty$  and  $\lim_{n \rightarrow +\infty} v_n = -\infty$ , then  $\lim_{n \rightarrow +\infty} (u_n + v_n) = -\infty$  ;

**(c) Product rule :**

- If  $\lim_{n \rightarrow +\infty} u_n = +\infty$  and  $\lim_{n \rightarrow +\infty} v_n = +\infty$ , then  $\lim_{n \rightarrow +\infty} (u_n \times v_n) = +\infty$  ;
- If  $\lim_{n \rightarrow +\infty} u_n = -\infty$  and  $\lim_{n \rightarrow +\infty} v_n = -\infty$ , then  $\lim_{n \rightarrow +\infty} (u_n \times v_n) = +\infty$  ;

**(d) Quotient rule :**

- If  $\lim_{n \rightarrow +\infty} u_n = +\infty$  and  $0 < v_n \leq M$ , then  $\lim_{n \rightarrow +\infty} \frac{u_n}{v_n} = +\infty$  ;
- If  $\lim_{n \rightarrow +\infty} u_n = +\infty$  and  $m \leq v_n < 0$ , then  $\lim_{n \rightarrow +\infty} \frac{u_n}{v_n} = -\infty$  ;
- If  $|u_n| \leq M$  and  $\lim_{n \rightarrow +\infty} v_n = \infty$ , then  $\lim_{n \rightarrow +\infty} \frac{u_n}{v_n} = 0$ .

## 6.3 Infinite limits and order relation

**Theorem 3.67** Let  $(u_n)_n$  and  $(v_n)_n$  be two sequences of real numbers , then

- If  $\lim_{n \rightarrow +\infty} u_n = +\infty$  and  $\lambda > 0$ , then  $\lim_{n \rightarrow +\infty} (\lambda u_n) = +\infty$  ;
- If  $\lim_{n \rightarrow +\infty} u_n = -\infty$  and  $\lambda > 0$ , then  $\lim_{n \rightarrow +\infty} (\lambda u_n) = -\infty$  ;

To be cont.